

Kenya Financial Inclusion Risk Prediction

Project Objective

1

Predict individuals at risk of financial exclusion in Kenya.

2

Use machine learning to support financial inclusion policies.

3

Provide transparent and explainable predictions.

4

Enable targeted interventions for vulnerable groups.

Business Problem



Millions remain outside formal financial systems.



Financial exclusion limits economic growth and resilience.



Manual identification of high-risk groups is inefficient.

Dataset Overview



- Source: Kenya FinAccess 2021 Survey.
- Contains demographic, socioeconomic and behavioral indicators.
- Features include education, age, location, employment and financial behavior.
- Target variable: Financially Excluded vs Included.

Data Preparation Pipeline

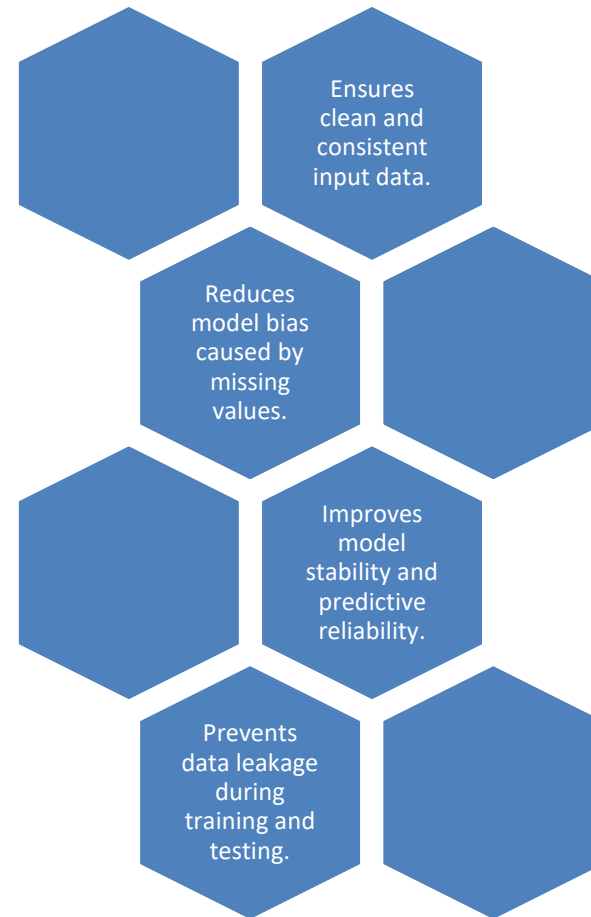
Missing categorical values filled using most frequent category.

Missing numerical values filled using median imputation.

OneHotEncoder transformed categorical variables.

StandardScaler normalized numerical features.

Why Preprocessing Matters



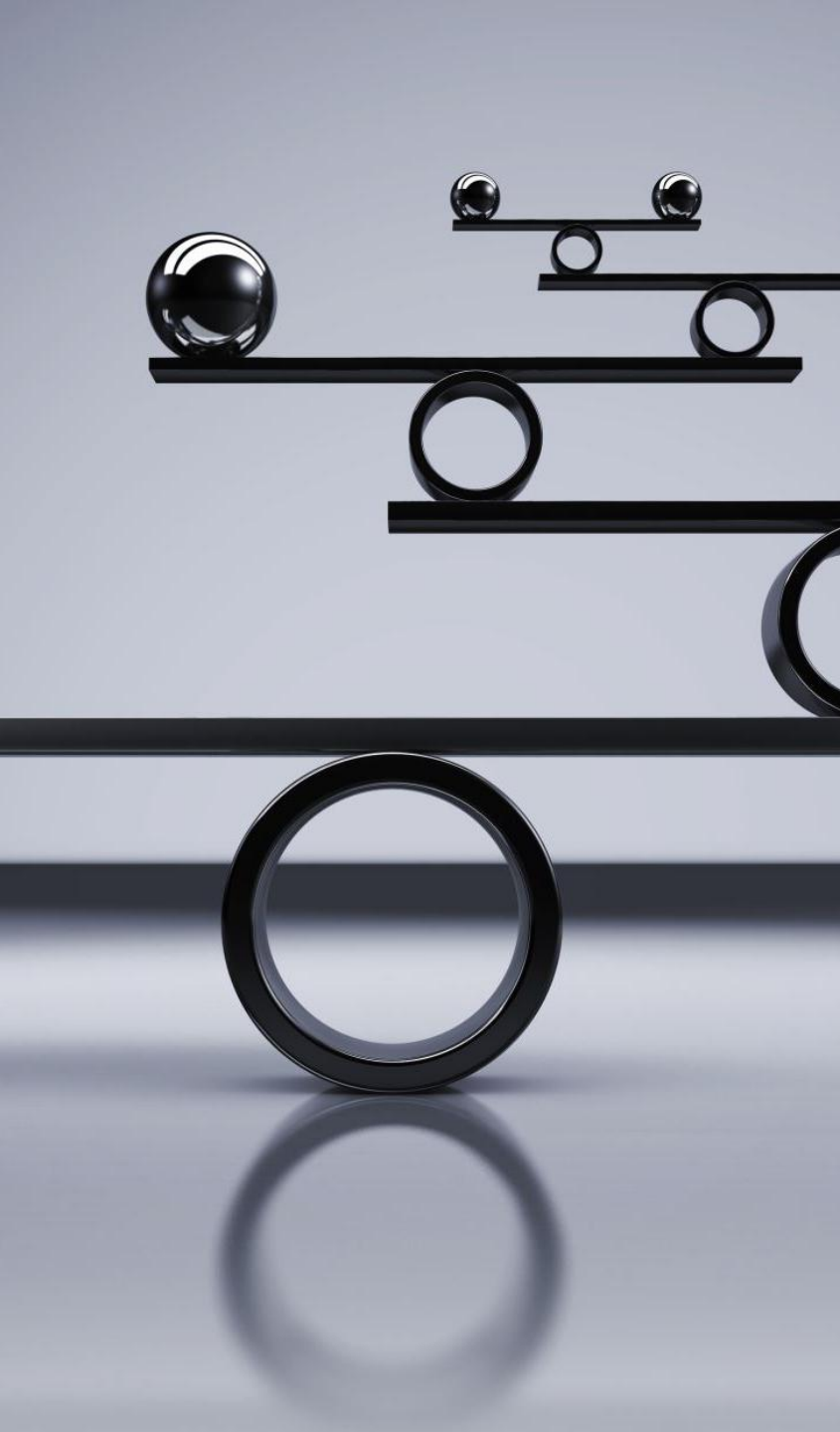
Models Evaluated

- Dummy Classifier: Baseline comparison model.
- Logistic Regression: Simple and interpretable classifier.
- Random Forest: Ensemble model capturing non-linear patterns.
- Gradient Boosting: Sequential boosting model optimized for performance.

Why Multiple Models Were Tested

- Different algorithms capture different data patterns.
- Linear models improve interpretability.
- Tree-based models improve complex pattern detection.
- Comparing models ensures robust final model selection.





Evaluation Metrics Explained

- Accuracy: Overall percentage of correct predictions.
- Recall: Ability to correctly identify excluded individuals.
- Precision: Reliability of positive predictions.
- F1 Score: Balance between recall and precision.
- ROC-AUC: Measures overall classification quality.

Understanding Model Trade-Offs



- High accuracy alone is not enough in imbalanced datasets.
- Recall was prioritized to avoid missing excluded individuals.
- False negatives are costly because vulnerable people may be ignored.
- Balanced metrics were used for fair model comparison.



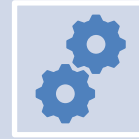
Best Model Selection

- Gradient Boosting achieved the strongest overall predictive performance.
- It captured complex nonlinear relationships in the data.
- Provided better balance between accuracy and generalization.
- Selected as final production-ready model for explainability analysis.

Why Gradient Boosting Performed Best



Builds trees sequentially to correct previous errors.



Handles mixed feature interactions effectively.



Reduces bias while maintaining strong predictive power.



Works well with structured socioeconomic survey data.

Model Explainability with SHAP



- SHAP explains how features influence predictions.
- Shows why an individual was classified as excluded.
- Improves transparency and trust .
- Supports policy interpretation and stakeholder confidence.

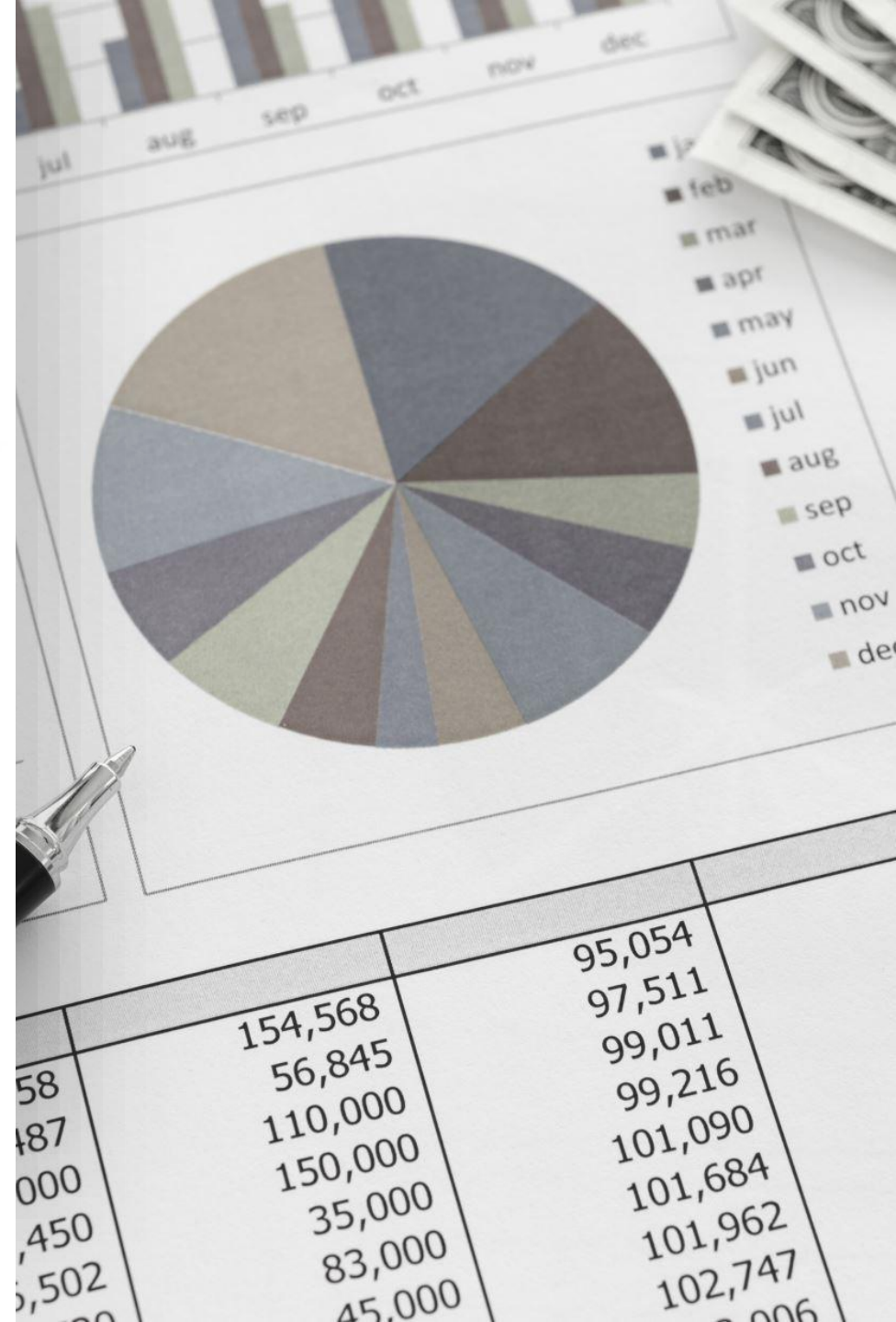
Key SHAP Insights

- Low education increased exclusion risk significantly.
- Rural residence strongly contributed to exclusion predictions.
- Youth populations showed higher vulnerability.
- Socio-economic status influenced financial inclusion outcomes.



Feature Importance Analysis

- Education level was among the strongest predictors.
- Employment and income indicators improved inclusion probability.
- Geographic location influenced access to financial services.
- Behavioral variables captured financial engagement patterns.

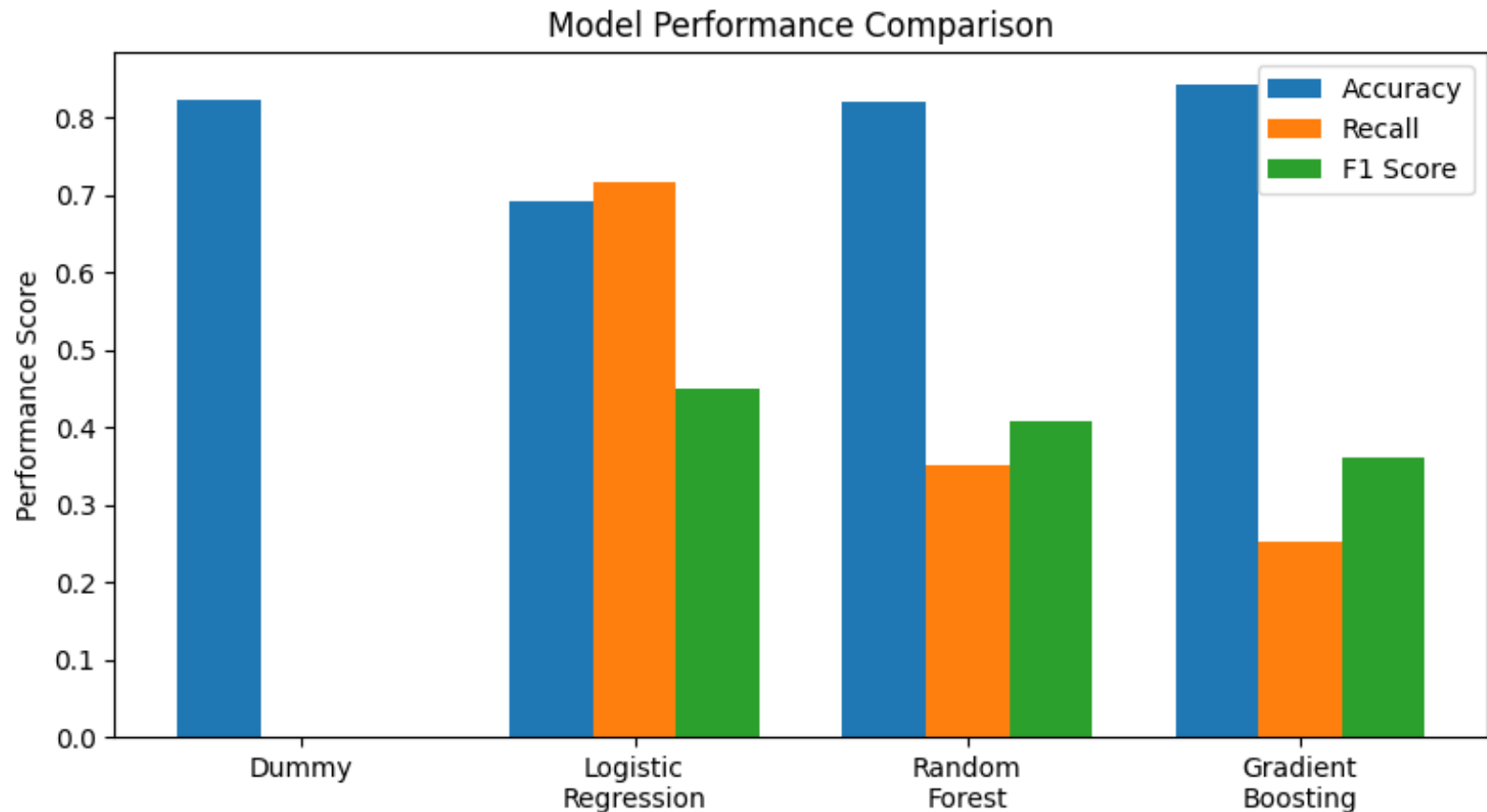


Project Impact

- Supports evidence-based financial inclusion policies.
- Can help banks target underserved populations.
- Improves transparency in predictive decision systems.
- Demonstrates responsible and explainable AI application in Kenya.



Model Performance Comparison



Deployment Overview

Analysis

Final Gradient Boosting model
deployed using Flask

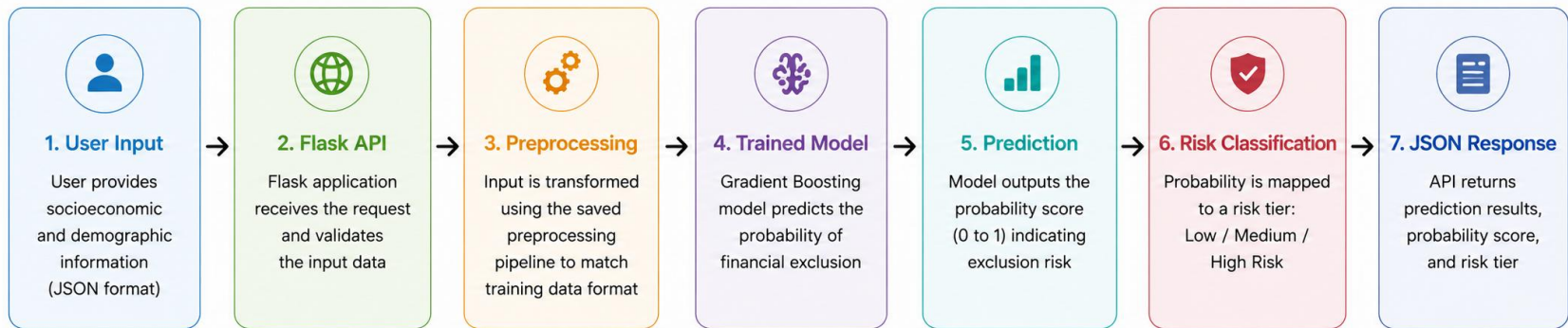
Saved model, preprocessing pipeline
and metadata for reproducibility

Enables non-technical users to
generate predictions

Demonstrates an end-to-end machine
learning workflow

Goal: Convert a trained model into a
practical decision-support application.





Deployment Architecture & Workflow

Core Assets:

- Trained model pipeline (.joblib)
- Feature schema (.json)
- Model metadata
- Sample input templates
- Ensures consistent preprocessing and reliable predictions in production.

Business Value & Impact

Risk Classification:

- Low Risk: <40%
- Medium Risk: 40–70%
- High Risk: >70%

Business Benefits:

- Accessible predictions for stakeholders
- Supports targeted financial inclusion programs
- Enables future dashboard and API integration
- Demonstrates production-ready deployment
- Takeaway: Deployment turns model outputs into actionable insights.

Conclusion

- Machine learning successfully identified financial exclusion patterns.
- Gradient Boosting delivered the strongest overall predictive performance.
- SHAP explainability improved model transparency and trust.
- The project demonstrates to improve and support financial inclusion in Kenya.





Kenya Financial Inclusion Risk Prediction

- Key Contributions
 - Defined a data-driven approach to identify individuals at risk of financial exclusion in Kenya
 - Cleaned, prepared and explored FinAccess data to uncover key patterns
 - Built and compared multiple machine learning models to predict exclusion risk
 - Applied SHAP, Mutual Information and Permutation Importance to explain predictions
 - Generated actionable recommendations for policymakers and financial institutions
 - Deployed a production-ready prediction system for decision support

Final Takeaway:

- "By combining data analysis, predictive modeling, SHAP and deployment, we developed an end-to-end solution to support financial inclusion in Kenya."

Thank You for Your Time